Jingwei Liu, Ph.D.

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Research Profile

Music AI researcher with a Ph.D. in signal and information processing and nearly seven years of R&D at Huawei. As project leader and system architect, led symbolic and audio generation and MIR systems deployed in Huawei Music, the Yinyuejia tablet DAW and smartphone Gallery. Current independent research focuses on MECO melody harmonisation.

Research interests: Controllable symbolic music generation; melody harmonisation; computer-assisted composition; MIR for music creation; musician-centred evaluation.

Professional Experience

Huawei Technologies Co., Ltd. Sep 2019 - Jun 2026

Most recent roles: AI Music Team Leader (PL), Principal Engineer (Level 18), and Music Creation System Architect.

Education

Ph.D. in Signal and Information Processing Sep 2014 - Jun 2019

University of Chinese Academy of Sciences; doctoral research at the Institute of Acoustics, Chinese Academy of Sciences, Beijing.

Supervisor: Prof. Jun Yang. Thesis: *Active Noise Control of Scattered Sound Fields Using Loudspeaker Arrays.*

B.Sc. in Physics, Fudan University Sep 2010 - Jul 2014

Selected Research: Music Generation

MECO Melody Harmonisation Jul 2026 - present

Independent research | Ongoing, unpublished prototype

- Developing an autoregressive Transformer for melody-conditioned chord generation. The public prototype supports MIDI import, audition and editable MIDI export: interactive demo.

Symbolic Music Generation Mar 2020 - Jan 2023

Huawei | Project Leader / Principal Engineer | Five-member AI music team

- Led a complete-song system combining melody, accompaniment, lyrics, singing synthesis and mixing, launched as Huawei Music's annual flagship feature in early 2023.
- Introduced chord and section controls, music-theory-guided sampling and sampling rollback to guide generation from limited, uneven-quality MIDI data.
- Used section-wise generation under global constraints on musical form, harmony and lyric metre to construct complete verse-chorus songs.

Full-Song Audio Generation Mar 2023 - Jun 2026

Huawei | Project Leader / Principal Engineer | Ten-member AI music team

- Led conditional full-song audio generation; a system from the programme was deployed in Huawei Music's 2024 New Year Campaign.
- Designed later codec-based Transformer generation with flow-matching audio rendering, and proposed representation alignment with pretrained music and speech encoders.
- Developed timbre, style and structure controls, supported by MIR pipelines for phrase segmentation, singing-voice ASR and audio-quality analysis.

Selected Research: MIR and Music Creation

MIR Algorithms for Huawei's Tablet DAW
Jul 2025 - Jun 2026

Huawei | System Architect / Technical Lead

- Designed and led chord recognition, source separation and singing-conditioned accompaniment algorithms for Huawei Yinyuejia, integrated into its music-editing workflow.
- Developed multi-style accompaniment from users' singing recordings, using pitch-shift and time-stretch augmentation to address imperfect pitch and timing.
- Combined MIDI transcription and phrase segmentation for song-level chord analysis. A Transformer chord recogniser achieved **83%** accuracy on basic triads and **64%** on seventh and ninth chords in internal evaluation.
- Improved source separation through targeted data augmentation, with internal SDR values of 18.9 dB (vocals), 21.85 dB (piano), 19.67 dB (bass) and 21.13 dB (drums).

Video Soundtrack and Music Recommendation
Jan 2024 - Jul 2025

Huawei | Project Leader / System Architect

- Led soundtrack generation and audio-visual alignment for Huawei's smartphone Gallery app, combining music, sound effects and selected original audio.
- Used beat detection, phrase segmentation and target duration to align music with video and guide the final mix.

Doctoral Research

Active Noise Control of Scattered Sound Fields
Sep 2014 - Mar 2019

Combined loudspeaker-array optimisation using constrained matching pursuit with zero-pressure constraint points to control the sound field scattered by a rigid sphere. Published as first author in *The Journal of the Acoustical Society of America* (2018).

Teaching, Mentoring and Leadership

- **Industry mentor for a postdoctoral researcher**, Huawei (Nov 2024 - Jun 2026): Guided research direction, methodology and project execution in music generation.
- **Technical mentor**, Huawei (2021 - 2026): Mentored six new employees in algorithm development and project execution.
- **Music theory lecturer**, Huawei (2024 - 2025): Designed and delivered a one-year internal lecture series attended by over 100 employees.
- **Technical interviewer**, Huawei (Apr 2023 - Jun 2026): Evaluated more than 100 AI audio candidates and contributed to four hires.

Musical Practice

Guitar, drums, accordion, keyboards and bass; years of band performance and original songwriting. Practical experience in harmony, composition, arrangement and production informs my work on music-creation tools.

Awards

President's Award for Technological Breakthrough, Huawei Central Media Technology Institute: 2024 (two achievements), 2023, and 2022 (two achievements).
President's Award, Huawei 2012 Laboratories, 2021. Internal corporate awards.

Peer-Reviewed Publications

Five publications: three journal articles and two conference papers.

1. R. Wang, X. Wang, **J. Liu**, and J. Yang. "Dual-zone active noise control algorithm." *Applied Sciences*, 10(1), 4, 2020. doi:10.3390/app10010004.

2. X. Wang, B. Liu, J. Yang, **J. Liu**, and R. Wang. "Utilizing vector hydrophones to achieve an active anechoic terminal in an acoustic tube." *Acta Acustica*, 44(2), 258-264, 2019. doi:10.15949/j.cnki.0371-0025.2019.02.013.

3. **J. Liu**, X. Wang, M. Wu, and J. Yang. "An active control strategy for the scattered sound field control of a rigid sphere." *The Journal of the Acoustical Society of America*, 144(1), EL52-EL58, 2018. doi:10.1121/1.5046446.

4. **J. Liu**, X. Wang, M. Wu, and J. Yang. "Actively created quiet zones using directional array of loudspeakers." *24th International Congress on Sound and Vibration (ICSV24)*, London, UK, 6181-6187, 2017.

5. X. Wang, B. Liu, J. Yang, Z. Jin, and **J. Liu**. "Adaptive active broadband absorption pipeline system design." *8th International Congress on Image and Signal Processing (CISP)*, 1525-1529, 2015. doi:10.1109/CISP.2015.7408126.

Patents and Patent Applications

Assignee: Huawei Technologies Co., Ltd.

1. **J. Liu** and Y. Kou. *Vlog Generation Method and Related Device.* Chinese patent CN 117156173 B, granted 2025. Application: CN 202210562029.3.

2. **J. Liu** et al. *Method for Generating Theme Sound and Electronic Device.* Chinese patent application CN 120856824 A. Application: CN 202410511744.3, filed 2024.

3. Z. Wang, M. Wang, C. Du, F. Fan, **J. Liu**, X. Li, and D. Xu. *Method and Apparatus for Processing Audio Data.* International patent application WO 2023/010949 A1, published 2023. PCT/CN2022/093923.

Selected Proprietary Technologies

J. Liu et al. *A Method for Multi-Track Music Generation Based on Flow Matching* (2026); *A Method for Autoregressive Music Generation Based on Discrete Codec* (2025). Huawei internal proprietary technologies.

Research Funding Experience

Huawei internal R&D (2020 - 2026): Project lead for music AI, responsible for proposals, roadmaps, resource coordination, and deployment.
NSFC-funded doctoral research: Core participant in projects on harmonic active noise control (11474306, 2014 - 2018), transfer-path analysis (11472289, 2014 - 2018), and active-passive sound absorption (11404368, 2014 - 2017).

Technical and Musical Skills

Programming: Python, PyTorch, C, MATLAB.
Modelling: Transformers, diffusion models, neural audio codecs, VAEs, GANs, reinforcement learning.
Audio and MIR: Music generation, chord recognition, structure analysis, source separation, music recommendation, audio mixing, sound-field control.
Music: Theory, harmony, form analysis, composition, arrangement and production; guitar, drums, accordion, keyboards and bass.
Languages: Mandarin Chinese (native); English (IELTS 7.5).